SANA AMANAT

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PROFESSIONAL SUMMARY

Motivated Computer Engineering student at COMSATS University Islamabad, Lahore Campus, with a strong foundation in programming, database systems and web development . Passionate about Artificial Intelligence, Generative AI, Cloud Computing, and emerging technologies. Experienced in developing academic and real-world projects using Python, Java, SQL, and web technologies. Strong problem-solving, teamwork, and leadership abilities demonstrated through technical projects and community engagement initiatives. Eager to join Summer Internship Program to learn from industry professionals, explore AI-driven solutions, and contribute to innovative technology projects.

EDUCATION

COMSATS University Islamabad, Lahore Campus — 2023 – Present
Bachelor of Science in Computer Engineering | CGPA: 3.62 / 4.00
Completed 6th Semester

Punjab College, Kasur — 2021 – 2023
ICS – Secured 81%

RELEVANT COURSEWORK

Programming Fundamentals • Object-Oriented Programming • Data Structures & Algorithms • Database Systems • Differential Equations • Linear Algebra • Digital Image Processing • Artificial Intelligence Fundamentals

AI & EMERGING TECHNOLOGIES EXPOSURE

- Studied Artificial Intelligence and Machine Learning concepts through academic coursework and self-learning.
- Worked on Digital Image Processing projects involving image enhancement, segmentation, and disease detection techniques.
- Familiar with Python for data analysis and AI-related applications.
- Exploring Generative AI tools, prompt engineering, and AI-assisted software development workflows.
- Interested in intelligent systems, automation, cloud technologies, and scalable software solutions.

SELECTED PROJECTS

American Sign Language (ASL) Alphabet Recognition System | Python, TensorFlow, Keras, CNN, Gradio

- Designed and trained a custom deep Convolutional Neural Network (CNN) to recognize all 26 ASL letters plus 3 special gestures (del, nothing, space) — 29 total hand-gesture classes.
- Trained on 43,500 high-resolution RGB images (1,500 per class) from the Kaggle ASL Alphabet dataset, achieving high accuracy on the held-out test set.
- Built three convolutional blocks with Batch Normalization and Dropout regularization from scratch using TensorFlow/Keras.
- Developed an interactive real-time web application using Gradio with a luxury dark aesthetic UI for image upload and prediction.
- Aligned with SDG 10 (Reduced Inequalities), SDG 4 (Quality Education), and SDG 9 (Industry, Innovation & Infrastructure) through accessible assistive AI technology.

Foliar Disease Detection System | MATLAB, Digital Image Processing

- Developed an image-processing-based system for automated plant leaf disease detection.
- Applied image enhancement, segmentation, and feature extraction techniques to identify disease patterns.
- Evaluated system performance using MSE and PSNR quality metrics.

- Improved understanding of computer vision and agricultural technology applications.

Sakura Corp Enterprise Network Design | Cisco Packet Tracer, OSPF, VLAN, ACL

- Designed and implemented a complete enterprise network for Sakura Corp spanning HQ and two regional branches (Branch A & Branch B) as Lead Network Architect.
- Configured VLAN segmentation, centralized DHCP and DNS services, and dynamic routing using OSPF.
- Implemented Router-on-a-Stick for inter-VLAN routing, Layer 2 security with port security and VTP.
- Applied Access Control Lists (ACLs) to enforce traffic policies between staff and guest users, ensuring secure and isolated inter-branch communication.

Cattle Farm Management System | MySQL, PHP, HTML, CSS, JavaScript

- Developed a real-world web-based management system with role-based access and operational tracking.
- Implemented database concepts including SQL joins, subqueries, views, procedures, and triggers.
- Designed interfaces for managing animals, milk records, users, and reports.
- Strengthened problem-solving, teamwork, and project development skills through full-stack implementation.

Hospital Management System | Java, Object-Oriented Programming

- Developed a Java-based application to manage patients, doctors, and appointments.
- Applied OOP concepts including classes, inheritance, and encapsulation.

TECHNICAL SKILLS

Programming Languages: C/C++, Java, Python, JavaScript, SQL
AI & ML Frameworks: TensorFlow, Keras, OpenCV (basic), MATLAB, Gradio
Web Technologies: HTML, CSS, PHP, JavaScript, React (Beginner)
Database Technologies: MySQL, SQL Server
Tools & Platforms: Git & GitHub, VS Code, Jupyter Notebook, Google Colab, MySQL Workbench, Gradio
Core Skills: Database Systems, Software Development, Web Development, Data Analysis, Computer Vision (Basic), AI Model Development, Problem Solving
Interests: Artificial Intelligence, Generative AI, Cloud Computing, Automation, Machine Learning

LEADERSHIP & EXPERIENCE

Female Ambassador – Women Development Department Sep 2025 – Jan 2026

- Represented and coordinated departmental initiatives focused on leadership and youth engagement.
- Developed communication, networking, and public relations skills through collaborative activities and events.
- Assisted in project coordination and teamwork-based community activities.

SOFT SKILLS

Communication & Collaboration • Leadership & Teamwork • Critical Thinking • Time Management • Adaptability • Community Engagement